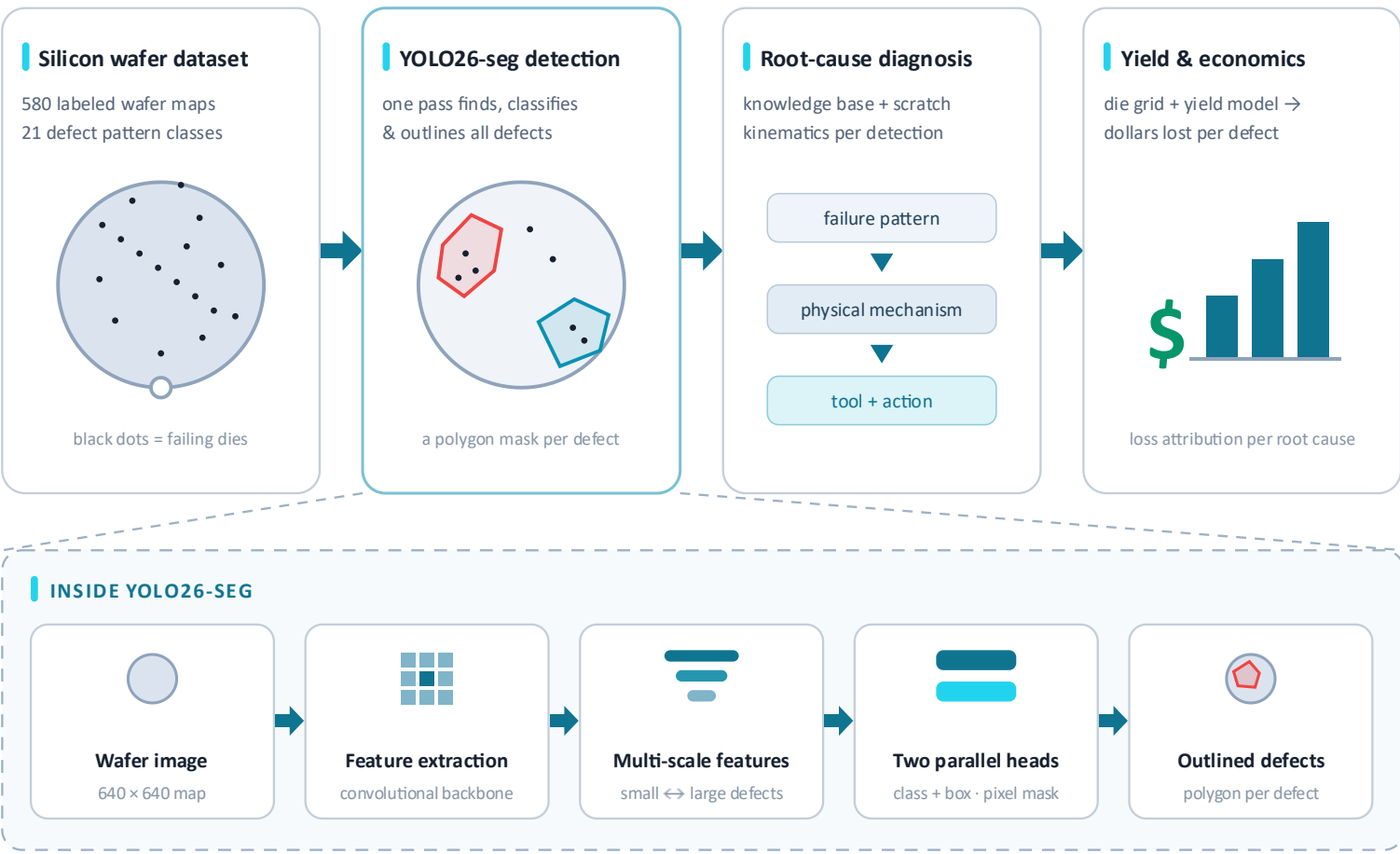


INTRODUCTION

Chips power everyday tech from phones and cars to AI tools, making semiconductor supply a major news issue. Semiconductor fabrication labs (fabs) build chips on silicon wafers through hundreds of steps. Every die (chip) on a wafer is electrically tested, and defective dies form a pattern linked to a root cause: scratches indicate handling damage, rings indicate polishing drift, slip lines indicate thermal stress, etc. With single wafers worth up to \$25k, misreading errors or failing to correct them is expensive. Yet prior works assign each wafer a single label with no location, no overlapping defects, and no way to measure cost.

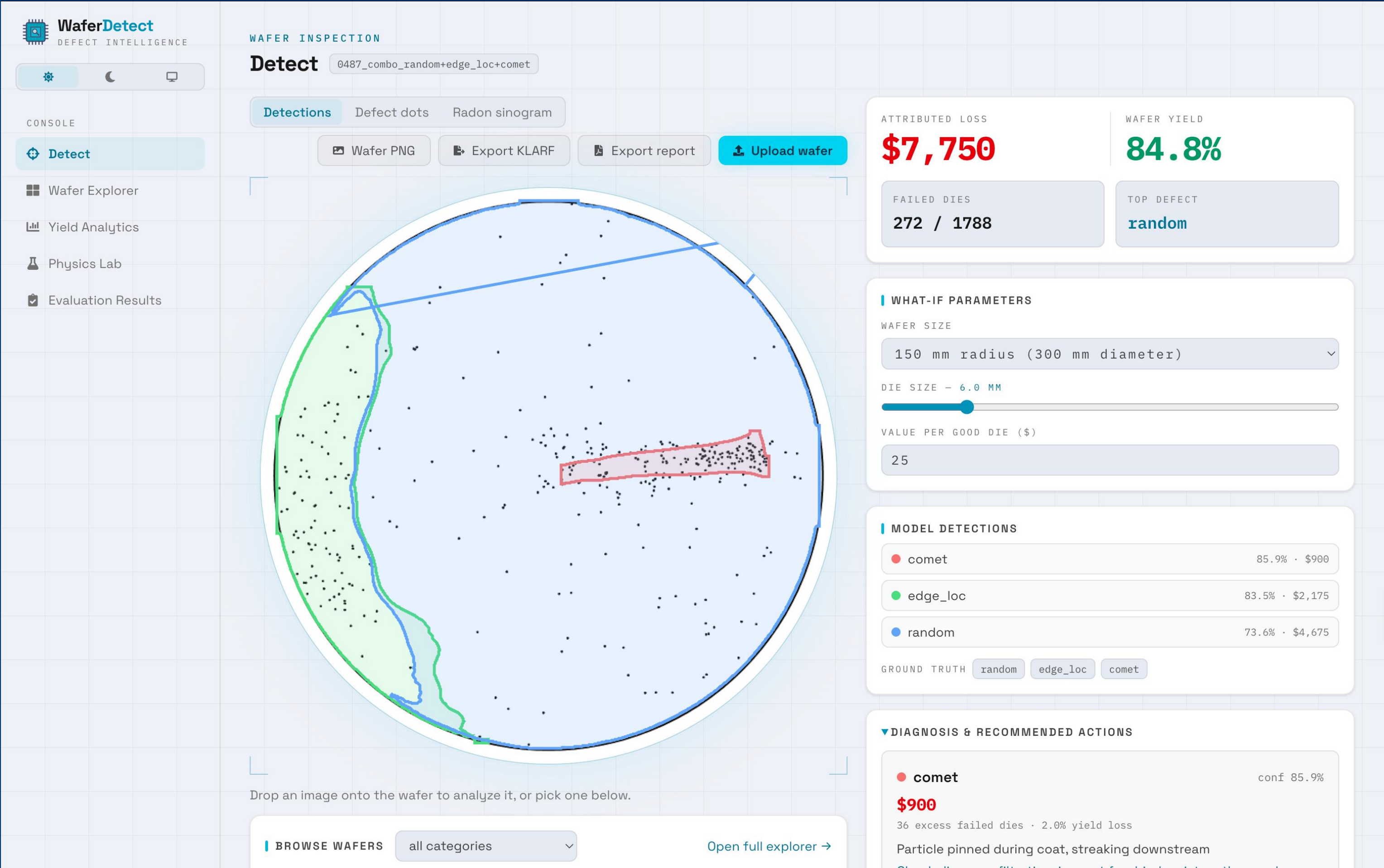
METHODOLOGY

- 1. **Dataset:** 580 annotated silicon wafer maps, spanning 21 defect pattern classes.
- 2. **Model Architecture:** YOLO26x segmentation model, a single-pass detector with a convolutional backbone that extracts features at multiple scales as input to parallel heads that predict each defect's class, box, and pixel mask together, outlining every defect.
- 3. **Analysis:** Each detection queries a root-cause knowledge base for the manufacturing error, recommended action, and the yield model used to quantify failed dies and dollars lost.



WaferDetect: Segmenting Complex Defect Patterns in Silicon Wafers with YOLO for Root-Cause Diagnosis

Rishab Alagharu



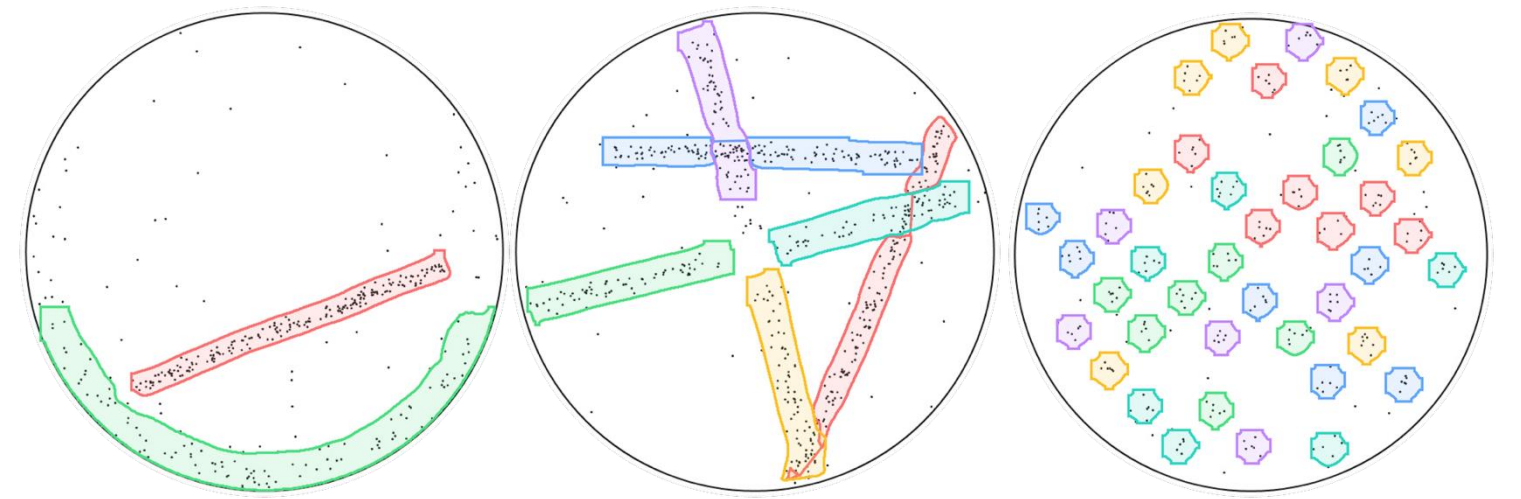
Scan to View Source Code



RESULTS

- For each silicon wafer map image, run the YOLO model and compute the Box and Mask Mean Average Precision (mAP).
- The test set consists of 87 wafers, containing all 21 classes.

	Box mAP	Mask mAP
Full Test	0.909	0.852
Combo	0.782	0.686
Edge Scratch Tiny	0.995	0.995



DISCUSSION & FUTURE WORKS

In a real fab line, automating the detection and cost analysis process can save fabs millions of dollars. Because the model is single-pass and runs in milliseconds, it can sit directly on a live, real-time wafer-test data stream, segmenting each wafer map as it comes off the prober, flagging errors across lots and alerting engineers to defects before they damage more wafers.

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